

Alexander Ricciardi

Computer science, software development, and applied AI/ML professional

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Profile

Computer science professional and Master of Science student in Artificial Intelligence and Machine Learning with a Bachelor of Science in Computer Science, Summa Cum Laude. Skilled in Python, Java, C++, TensorFlow, PyTorch, RAG, vector search, knowledge graphs, AI agents, and LLM application development, with hands-on experience building documented software projects and technical field-service experience since 2008. Seeking software engineering internships focused on applied AI and scalable intelligent systems.

Professional interests and subject areas

Software engineering and development; AI engineering; AI agents and agentic systems; retrieval-augmented generation (RAG); vector search, VectorRAG, GraphRAG, and knowledge graphs; LLM application development and model/API integration with OpenAI, Anthropic, DeepSeek, and other providers; artificial intelligence and machine learning; data analysis and business intelligence; AI research; Python, Java, C++, SQL, databases, data structures and algorithms; technical instruction, adjunct teaching, bootcamp instruction, employee AI training, generative AI and chatbot use, and responsible technology use.

Education

Master of Science, Artificial Intelligence and Machine Learning <i>Colorado State University Global</i>	March 2027
Bachelor of Science, Computer Science <i>Colorado State University Global</i>	August 2025
• Achievements: Summa Cum Laude • Coursework: Operating Systems and Architecture, Software Engineering, Database Management, Networks, Cybersecurity, Java, Python, C++, Artificial Intelligence, Data Structures and Algorithms	
Associate of Science, Computer Science <i>Laramie County Community College</i>	December 2023
• Achievements: High Distinction Honors	
Certificate of Completion, Full Stack Web Development <i>The Array School of Technology and Design</i>	June 2018

Academic projects

CubeSat telemetry anomaly detection Colorado State University Global	2026 - Present
• CSC510 course AI portfolio project, Colorado State University Global • Implemented a Python decision-support program using neural-network classification and A* search to analyze CubeSat telemetry anomalies.	
Mining Regulatory Compliance Assistant	2025
• Bachelor of Science capstone project • Built a mine regulatory AI-powered assistant that queries Mine Safety and Health Administration regulations. • Combined VectorRAG and GraphRAG retrieval with context fusion to build an advanced parallel hybrid RAG system using Python, FastAPI, Neo4j, Streamlit, and Docker.	
Public computer science academic portfolio	2022 - Present
• GitHub portfolio repo. showcasing code, papers, and projects. • Publicly provides coursework and projects in Python, Java, C++, artificial intelligence, algorithms, operating systems, databases, networks, software engineering, cybersecurity, Android development, and graphics.	

Professional experience

Rocky Mountain Business Equipment Copier Technician	2008 - Present
Technical field service since 2008	
• Apply engineering principles to diagnose and resolve complex equipment problems while repairing, performing preventive maintenance on, and installing office equipment for business customers. • Work with business customers to determine service requirements and maintain Kyocera copiers, printers, facsimiles, scanners, and related multifunction systems.	

Master's Progress

Course | Grade
CSC510 Foundations of Artificial Intelligence | In progress
CSC507 Foundations of Operating Systems | A / 4.0
CSC506 Design and Analysis of Algorithms | A / 4.0

CSC505 Principles of Software Development | A / 4.0

CSC502 Ethical Leadership in Software Development | A / 4.0

CSC501 Management for the Computer Science Professional | A / 4.0

CSC500 Principles of Programming, Python | A / 4.0

Skills

- **Programming:** Python, Java, C++, Kotlin, JavaScript, SQL, HTML, CSS
- **AI and data:** TensorFlow, PyTorch, NumPy, Pandas, Jupyter, Matplotlib, Plotly, neural networks, retrieval-augmented generation (RAG), VectorRAG, GraphRAG, vector search, knowledge graphs, AI agents, LLM application development, machine learning, machine learning model evaluation, data science, retrieval systems, large language models, prompt engineering
- **Development:** FastAPI, Streamlit, Docker, Neo4j, MySQL, Git, GitHub
- **Academic communication:** Technical reports, APA research writing, public code examples, project documentation

Links

Omegapy GitHub. github.com/Omegapy - Software, AI, and academic projects.

Academic portfolio. github.com/Omegapy/My-Academics-Portfolio - Coursework, source code, papers, and project records.

AngryOwlAI. github.com/AngryOwlAI/ - AI project organization.

Code Chronicles. alexomegapy.com - Personal technical blog.

Medium. medium.com/@alex.omegapy - Technical articles and commentary.

Languages

- English(fluent)
- French(fluent)
- Spanish(fluent)
- Italian(spoken)